



New Year Quiz

1. What best describes Motherboard's core innovation in interoperability?

- A. A faster trust-minimized cross-chain bridge
- B. A brand-new generalized Layer-1 blockchain
- C. A modular mental model commoditising communication and distribution
- D. A unified globally shared relay infrastructure

Answer: C

2. Which actor in the Motherboard architecture is responsible for gas management across chains?

- A. Independent cryptographic verifier networks
- B. Decentralised transaction relay networks
- C. Read-write blockchain RPC service providers
- D. Core routing smart contract modules

Answer: B

3. What does “non-discriminatory distribution” specifically enable?

- A. Exclusive chain partnerships and integrations
- B. Permissioned deployment across approved networks
- C. Fast cheap permissionless scaling to new chains
- D. Validator reward alignment and optimisation

Answer: C

4. When is a transaction officially considered valid on the destination chain?

- A. Once the message payload is delivered
- B. When cross-chain gas fees are settled
- C. After off-chain client execution completes
- D. When verifier report and relay signature both match

Answer: D

5. How does Motherboard specifically expand Total Addressable Market (TAM) for applications?

- A. By reducing deployed smart contract complexity
- B. By enforcing protocol-wide standard interfaces

- C. By acting as a neutral distribution rail across chains
- D. By limiting deployment to approved ecosystems

Answer: C

6. Which definition best captures “messaging” as described in Andy Bohutsky’s article?

- A. Broadcasting transactions simultaneously to all chains
- B. Copying global blockchain state across networks
- C. Passing instructions or data so events trigger across chains
- D. Synchronizing block production times across ledgers

Answer: C

7. What problem did the “hub and spoke” interoperability model attempt to solve?

- A. Validator power concentration across networks
- B. Fragmentation of users and liquidity across chains
- C. Excessive gas fees on base layer networks
- D. Increasing smart contract development complexity

Answer: B

8. The earliest interoperability model (circa 2017) is compared to which telecommunication system?

- A. Early consumer internet broadband networks
- B. Modern cellular communication infrastructures
- C. Telegraph-style point-to-point communication
- D. Long-range satellite relay communication

Answer: C

9. What is the correct order of steps for a message to move from chain A to chain B?

- A. Deliver message, read event, validate proof, write state
- B. Read event, validate proof, deliver message, write state
- C. Validate proof, read event, write state, deliver message
- D. Read event, deliver message, validate proof, write state

Answer: B

10. Why are blockchains described as monolithic by design?

- A. They rely on fully centralized validator committees
- B. They prioritise execution speed over security
- C. They only support a single programming language
- D. They cannot natively communicate with external systems

Answer: D

11. According to Andy Bohutsky's article, what is the first clear signal that interoperability is not solved?

- A. Persistently high transaction fees across chains
- B. Insufficient tooling available for developers
- C. Discriminatory distribution across blockchain ecosystems
- D. Slow and inconsistent transaction finality times

Answer: C

12. Why does the article argue that apps and asset issuers are "second-class citizens"?

- A. They lack the required technical infrastructure
- B. Their distribution is controlled by infrastructure providers
- C. They rely on chains to access liquidity
- D. They are unable to deploy across chains

Answer: B

13. Which statement best summarizes the current power dynamic in interoperability?

- A. Users dictate standards across all protocols
- B. Chains dictate how applications must behave
- C. Infrastructure dictates terms to value generators
- D. Value generators dictate infrastructure design

Answer: C

14. Which principle is emphasized as essential for the future of interoperability?

- A. Infrastructure ownership and value capture
- B. Permissioned access for verified participants
- C. Provider-centric standards and governance
- D. Neutral open trustless and permissionless systems

Answer: D

15. What percentage of the onchain world currently receives interoperability services from the largest interop protocol?

- A. Roughly half of existing blockchain networks
- B. About one quarter of deployed chains
- C. Approximately ten percent of the ecosystem
- D. Nearly one percent of all blockchains

Answer: C

16. What mechanism is used by interoperability providers to build moats?

- A. Highly restrictive validator admission requirements
- B. Carefully curated exclusive ecosystem partnerships
- C. Vendor-locking technical standards and interfaces
- D. Centralized governance-driven protocol decision making

Answer: C

17. Approximately how many EVM chains exist, according to the article?

- A. Roughly five hundred active EVM chains
- B. Nearly eight hundred deployed EVM chains
- C. About one thousand operational EVM chains
- D. Over one thousand four hundred forty-five EVM chains

Answer: D

18. Why are cell-tower-like interoperability models still considered problematic?

- A. They lack fully permissionless chain onboarding
- B. They cannot scale efficiently beyond Layer 2s
- C. They rely on rent-seeking and value extraction
- D. They remove validators from transaction verification

Answer: B

19. What ensures Motherboard remains trustless and neutral?

- A. A central coordinating authority overseeing transactions
- B. Fixed infrastructure providers controlling execution flow
- C. Independent actors with no single source of truth
- D. Permissioned participation governed by protocol rules

Answer: C

20. What input does the verifier network require to generate a tamper-proof report?

- A. User-defined write RPC endpoint address
- B. Cryptographic relay network signature proof
- C. User-defined read RPC endpoint address
- D. Destination blockchain unique chain identifier

Answer: C